RW-6068 **502101**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

MATHEMATICS AND BIOSTATISTICS

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define Sample Space.

2. Explain Systematic Sampling.

3. Write short note on Class-intervals.

4. Find the value of $1 - \sin^2 30^\circ$

5. Differentiate $\cot^{-1}(\log x)$

6. What is range ?

7. If $A = \begin{bmatrix} 5 & 1 \\ 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 2 \\ 3 & 4 \end{bmatrix}$, calculate

$B^2 + AB + 6A + A^2$.

8. Define Standard deviation.

9. Find the value of $1 - \sin^2 30^\circ$

10. Explain : Histogram.

Part - B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Define and differentiate mean, median and mode with example.

(*Or*)

(b) Give an account on Binomial distribution.

12. (a) Solve $\sqrt{1-x^2}\, dy + \sqrt{1-y^2}\, dx = 0$

(*Or*)

(b) If $y = e^{\sqrt{\tan x}}$ find $\frac{dy}{dx}$.

13. (a) Find a_o and a_n for the Fourier series expansion of $(x + x^2)$ in the interval $(-\pi, \pi)$.

(*Or*)

(b) Evaluate $\int 3x^2 e^x \, dx$.

14. (a) Comment on Chi-squared distribution and its applications

(*Or*)

(b) Calculate Arithmetic mean.

Class intervals	0—10	10—20	20—30	30—40	40—50
Frequency	6	5	8	15	7

15. (a) Distinguish between Mean deviation and Standard deviation.

(*Or*)

(b) Explain range and inter quartile range with suitable examples.

Part - C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the Lorenz curve and its significance.

17. Find $\frac{dy}{dx}$, $x^y = y^x$.

18. Give a detailed account on student's "t" distribution.

19. For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 2 & 2 \\ 1 & 0 & 2 \end{bmatrix}$ find A^{-1}.

20. List out the various Sampling methods.

——— *** ———

RW-6069 **502102**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

FUNDAMENTALS OF COMPUTING

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.

1. Expand :

   (a) PROM.

   (b) EEPROM.

2. What is a PDA ?

3. Define : Application software.

4. What is meant by Effective access time ?

5. What is a concentrator ?

6. Define the term, “protocol”

7. What is a Router ?

8. Expand

   (a) URL.

   (b) HTML.

9. List down the types of computer crimes.

10. Distinguish Encryption from Decryption.

Section - B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write about microcomputers.

Or

(b) Give a note on supercomputers.

12. (a) Narrate the features of LINUX operating system.

Or

(b) Compare and contrast system software with application software.

13. (a) Explain any two topologies of computer networks.

Or

(b) Give a note on ISDN architecture.

14. (a) With sample code, explain the role of HTML in internet.

Or

(b) Discuss on Web browsers and Search engines.

15. (a) Write about Grid computing.

Or

(b) Summarize the concept of authentication and cryptography.

Section - C (3 × 10 = 30)

Answer any **three** questions.

16. Write detailed notes on evolution of operating system.

17. Describe the device management policies.

18. With neat diagram, explain the OSI reference model.

19. List various internet protocols and explain them.

20. Explain the following concepts :

(a) Computer viruses and virus vaccines (5)

(b) Firewalls and protection mechanism. (5)

——— *** ———

RW-6070 **502103**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

CELL BIOLOGY

(CBCS-2010 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.
All questions carry equal marks.

1. Karyology.

2. Suicidal bag.

3. Trilamnar model.

4. Gap Junction.

5. Cellulose.

6. Plasodesmata.

7. Define : Apoptosis.

8. Giant chromosomes.

9. Biological clock.

10. Mutagens.

Part - B (5 × 5 = 25)

Answer **all** questions.

All questions carry equal marks.

11. (a) Brief the structure of endoplasmic reticulum.

(*Or*)

(b) Describe the structure of Golgi complex.

12. (a) Explain the process of phagocytosis.

(*Or*)

(b) Narrate the chemical composition of plasma membrane.

13. (a) Write about the thickenings in the cell wall.

(*Or*)

(b) Write about the tight junction.

14. (a) Give brief account on protein secretion.

(*Or*)

(b) Give a brief account on Chaperons.

15. (a) Write a note on tumor progression.

(*Or*)

(b) List out the changes occur during Aging.

Part - C

(3 × 10 = 30)

Answer any **three** of the following.

16. Compare and contrast the Prokaryotic cell and Eukaryotic cell.

17. Give an account on molecular basis of cell cycle.

18. Present the electron microscopic structure of cell wall.

19. Describe the process and mechanism of apoptosis.

20. Give an account on oncogenic viruses.

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RW-6071 **502104**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

LAB–I—ANALYTICAL AND INSTRUMENTATION METHODS

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.
All questions carry equal marks.

1. Write the names of light sources in a colorimeter.

2. State common features of colorimeter.

3. What is Km ?

4. State clinical significance of SGPT.

5. Write any four solvents used in the paper chromatography.

6. Define : R*f* factor.

7. Define : Dialysis.

8. Comment on ultrafiltration.

9. What is tracker dye ?

10. State the importance of Radio labeling.

Part - B (5 × 5 = 25)

Answer **all** questions.

All questions carry equal marks.

11. (a) Write down the principle and mechanism of a colorimeter.

Or

(b) Describe about any *four* components of a spectrophotometer.

12. (a) Bring out procedure for the assay of SGOT.

Or

(b) List down the clinical significance of lipases and ALP.

13. (a) Write about the calomel electrode of a pH meter.

Or

(b) Tabulate the suitable pH indicators for different types of titration.

14. (a) Write a brief note on the choice of buffers for Ion exchange chromatography.

Or

(b) Write about the ammonium sulphate precipitation.

15. (a) Explain the methods to measure the radio-activity.

Or

(b) Give a brief account on the factors affecting electrophoretic mobility.

Part - C (3 × 10 = 30)

Answer any **three** of the following

16. Narrate the Lowry's method of protein estimation.

17. Describe the estimation of blood glucose level. Add a note on importance of it at fasting, post prandial and random.

18. What is the Henderson Hasselbach equation ? Add a note on the application of Buffers in biological system.

19. What are the extra cellular enzymes ? Describe the production of amylase by using a fungus.

20. Give an account on the principle and working mechanism of tube gel electrophoresis.

——— *** ———

RW-6072 **502105**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

LAB II —PROGRAMMING IN C

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define : Conditional operators.

2. What will be the output of the following program segment ?

```
# include < stdio.h >

void main ( )

{

int a, b = 0 ;

int c [10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 0} ;
```

```
for (a = 0 ; a < 10 ; + + a)

b + = c [a] ;

printf (“ % d” , b) ;

}
```

3. Define an Array.

4. Write the difference between call-by value and call-by-reference.

5. Define : Pointer.

6. What is union ? How is it declared ?

7. Define : Pixel.

8. What is Transformation ?

9. How protein sequences are extracted from databank file ?

10. What is FASTA format ?

Part - B $(5 \times 5 = 25)$

Answer **all** questions by choosing **either** (a) **or** (b).

11. (a) Write a C program to find the largest of three numbers.

Or

(b) Explain the various loop structures available in C.

12. (a) Write a C program to convert the number of years represented by an integer into the following units of time :

(i) Days

(ii) Months

(iii) Hours.

State the assumption made in your problem.

Or

(b) Differentiate between Array of pointers and Pointers to array, with example.

13. (a) Write a C program that computes the addition of two matrices.

Or

(b) Explain storage classes in C.

14. (a) Write a program to draw the Circle.

Or

(b) Explain the different types of projections.

15. (a) Write a C program to convert FASTA format to PIR format.

Or

(b) How to locate palindromic sequence in a DNA ?

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Write a C program to print the given number in reverse order.

(5)

(b) Write a C program to check if a given number is a prime number.

(5)

17. Write a C program to read and write the details of ten employees.

18. Write a C program to find the largest and lowest number in an array.

19. Explain the various types of clippings.

20. Write a C program to count the number of Open Reading Frames in a DNA sequence.

——— *** ———

RW-6073 502503

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

BIOINFORMATICS IN AGRICULTURE, MEDICINE, BIODIVERSITY AND ENVIRONMENT

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.
All questions carry equal marks.

1. What is meant by cDNA ?

2. Define Drought resistance.

3. Define the term Preventive medicine.

4. What is the difference between Germline and Somatic gene therapy ?

5. What is meant by ecological diversity ?

6. What are the uses of virtual library ?

7. Define Superbugs.

8. Define Fuel cells.

9. What is drug resistance ?

10. Define Genome.

Part - B (5 × 5 = 25)

Answer **all** questions choosing *either* (a) *or* (b).

11. (a) List out any two Indian medicinal plants and explain its importance.

(*Or*)

(b) Describe Genetic diversity.

12. (a) Write a brief note on the various bio applications of gene therapy.

(*Or*)

(b) Write a short note on bacterial drug targets.

13. (a) Explain the scope of Biodiversity.

(*Or*)

(b) Describe special interest networks.

14. (a) Explain the various applications of fuel cells.

(*Or*)

(b) What is the difference between renewable and non-renewable energy ?

15. (a) Discuss on ortholog, paralog and xenolog.

(*Or*)

(b) Explain the various biotechnological applications of microbes.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. How to improve the nutritional quality and explain in detail ?

17. The aberrant expression of BCl - 2 and BCl-xl proteins result in the inhibition of apoptosis which becomes the major cause of the cancers. Nucleic acid lengths of these proteins were nearly 1207 bp in length. You are asked to treat these changes using the gene therapy mechanism. Sketch out the mechanism and explain in detail the steps involved in treating the disorders.

18. Write a detailed notes on Bibliographies.

19. Explain the relationship between microbes and climate change.

20. How to classify the molecular phylogenetic methods and explain in detail ?

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RW-6074 502301

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

COMPUTATIONAL APPROACHES TO TAXONOMY AND PHYLOGENY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.

1. Define Taxa.

2. Define Molecular Evolution.

3. What is Cladogram.

4. Define Phylip.

5. Define Boostrap analysis.

6. Define Dendrogram.

7. What is an Evolutionary Relationship ?

8. Define OTU.

9. Define READSEQ.

10. Define Binary Tree.

Section - B (5 × 5 = 25)

Answer **all** the questions choosing *either* (a) *or* (b).

11. (a) Explain the role of molecular evolution in the phylogeny analysis.

(*Or*)

(b) Explain the steps in constructing a phylogeny tree.

12. (a) How can the morphological data be characterized in phylogeny ?

(*Or*)

(b) How can the unrooted tree be constructed from the sequence data ? Explain.

13. (a) Explain the representation of phylogenetic tree.

(*Or*)

(b) Give an account of data partitioning and combination.

14. (a) How will one get the reliable phylogenetic tree ? Explain.

(*Or*)

(b) Write the applications of phylogeny analysis.

15. (a) Write the Dynamic programming of multiple sequence alignment.

(*Or*)

(b) How can multiple datasets be generated by using bootstrapping tool ? Explain.

Section - C (3 × 10 = 30)

Answer any **three** questions.

16. Explain briefly on taxonomic information on plants.

17. How can the phylogenetic tree be constructed based on characters ? Explain with an example.

18. Compare the phylogenetic trees obtained from DNA sequence *Vs* Protein sequences.

19. Illustrate about Clustal W algorithm and Phylip package.

20. Discuss in detail about reconstruction of phylogenetic tree using molecular data.

——— *** ———

RW-6075 **502302**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

COMPARATIVE FUNCTIONAL GENOMICS

(CBCS–2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** the questions.

Explain :

1. SNPs.

2. Genome.

3. Paralogs.

4. Open Reading Frame (ORF).

5. M-LAGAN.

6. Genome database.

7. miRNA.

8. Tiling array.

9. Microsatellite sequences.

10. Homologous recombination.

Part - B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Difference between Archaea and Eucarya genome.

(*Or*)

(b) Short note on arrangements of genes on chromosomes.

12. (a) Define Polymorphism and its types.

(*Or*)

(b) Give an account of NCBI map viewer.

13. (a) Short note on Genome annotation.

(*Or*)

(b) List out the applications of DNA microarrays.

14. (a) Short note on Repeats and Masking.

(*Or*)

(b) Give an account of SAGE.

15. (a) Write a brief note on Celera genomic project.

(*Or*)

(b) Describe the importance of inversion and translocation in genome mapping.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the structure of chromatin and its packaging.

17. Give an account of Physical and Genetic mapping.

18. Explain in detail about the bioinformatic tools for genomics and proteomics.

19. Give a detailed account on gene knock–out analysis and its limitations.

20. Describe in detail about the mechanism of antisense RNA to block the translation.

——— *** ———

RW-6076 | 502303

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

PROTEOMICS AND SYSTEMS BIOLOGY

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.

1. Define an amino acid.

2. Draw the structure of proline.

3. What are polar amino acids ?

4. Define Proteome mining ?

5. Define GOR.

6. Define Networks.

7. Define Rasmol.

8. What is HSSP ?

9. Define Micro array.

10. Define Template sequence.

Section - B (5 × 5 = 25)

Answer **all** questions.

11. (a) How to identify the components of a complex mixture ? Explain.

(*Or*)

(b) Explain the method of protein sequencing.

12. (a) Explain the classes of protein structure.

(*Or*)

(b) Write the features of CATH.

13. (a) Discuss about the protein structure prediction softwares.

(*Or*)

(b) Give an account of network structure and dynamics.

14. (a) Describe about *ab-initio* method of structure prediction.

(*Or*)

(b) Explain the properties of protein-protein complexes.

15. (a) Give an account of SARF database.

(*Or*)

(b) Explain the steps involved in homology modeling.

Section - C (3 × 10 = 30)

Answer any **three** questions.

16. Enumerate the method of genome sequence analysis by mass spectrometry.

17. Explain the following :

    (a) FSSP.

    (b) HSSP.

    (c) MMDB.

    (d) SCOP.

18. How will you predict the secondary structure of proteins ? Explain.

19. What are neural networks ? Explain the method involved to construct the model.

20. Give a account on structures of regulatory networks.

——— *** ———

RW-6077

502304

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

RECOMBINANT DNA TECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** the questions.

1. Can exonucleases act as restriction enzymes ? Why ?

2. Site the difference between *E.Coli* DNA ligase and T4 DNA ligase.

3. What are the selectable markers for Lamda vectors ?

4. Comment on Gene fusion vectors.

5. Explain labeling by Nick translation.

6. What are the chemical agents used to modify DNA for Maxam and Gilbert method ?

7. Briefly explain PCR cycle.

8. Give the applications of site directed mutagenesis.

9. Define the term DNA profiles.

10. Write short notes on Adenosine Deaminase (ADA) deficiency.

Part - B (5 × 5 = 25)

Answer **all** the questions.

11 (a) Write short notes on (i) Terminal transferases (ii) S1 nucleases (iii) Alkaline phosphatase.

(*Or*)

(b) What are Restriction enzymes ? Write about their Nomenclature and about the different classes of Restriction enzymes.

12. (a) Explain the properties of mp 18 and mp 19 vectors.

(*Or*)

(b) Compare Prokaryotic and Eukaryotic expression vectors.

13 (a) Explain in general about the different Gene cloning methods.

(*Or*)

(b) Write in detail on Western blotting and its applications.

14 (a) Compare Maxam and Gilbert and Sanger's dideoxy sequencing method.

(*Or*)

(b) Discuss on site-directed mutagenesis and its applications.

15 (a) Write an essay on the applications of rDNA technology in producing Recombinant proteins.

(*Or*)

(b) Role of Recombinant DNA technology in the field of Diagnostics.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. What are DNA modifying enzymes ? Explain in detail the role of any two enzymes in rDNA technology.

17. Explain in detail on different types of Artificial chromosomes.

18. Describe in detail the construction, screening and selection of Genomic DNA Library.

19. Write in detail about the Sanger's chain termination method.

20. What is RFLP ? How is it used in DNA Fingerprinting ?

——— *** ———

RW-6078

502305

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

Lab V—TEXT AND DATAMINING OF BIOLOGICAL SYSTEMS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

Define :

1. Core text mining.

2. Write any two applications of text mining.

3. AcroMed.

4. Profile analysis.

5. PRINTS.

6. Optimization.

7. Newton Raphson.

8. Adaline network.

9. Promoter prediction.

10. EM algorithm.

Part - B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Discuss in short about text mining pre-processing techniques.

(*Or*)

(b) Explain the following tools : AcroMed, MedMiner.

12. (a) Give a short note on profile analysis.

(*Or*)

(b) Outline the uses of pattern databases.

13. (a) Develop a 'C' program for epitope prediction.

(*Or*)

(b) Give an account on optimization algorithm.

14. (a) Explain about epitope prediction.

(*Or*)

(b) Implement the 'C' program for perception learning rule.

15. (a) Give short notes on hidden Morkov model implementation with C program.

(*Or*)

(b) How will you determine the entropy in DNA sequence ?

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Write a note on information extraction and categorization.

17. Write about the advanced techniques of sequence analysis.

18. Explain the implementation of steepest descent and conjugate gradient for energy minimization along with C program.

19. Implement the learning algorithm in promoter prediction.

20. Write an algorithm for determining the relative entropy in two sequence.

——— *** ———

RW-6079 **502306**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

Lab VI—MOLECULAR MODELING AND GRAPHICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section - A (10 × 2 = 20)

All the questions carry equal marks.

1. Define an internal co-ordination system.

2. What is called energy surface ?

3. What is an internet ?

4. Define Potential energy.

5. Define Molecular mechanics.

6. Define Torsion angle.

7. Define GPU.

8. What is lighting ?

9. Draw the structure of serine.

10. Define 3D-object.

Section - B (5 × 5 = 25)

Answer **all** questions choosing **either** 'a' **or** 'b'

11. (a) Explain the features of molecular mechanics force fields.

Or

(b) Enumerate the parameters of transferability force field.

12. (a) Give an account of Molecular mechanics and Quantum mechanics.

Or

(b) Define Coordinate systems. Explain with an example.

13. (a) What are the time-dependant properties of molecular dynamics ?

Or

(b) Write the applications of energy minimization.

14. (a) Illustrate about frame buffer techniques in graphics.

Or

(b) What are the interaction devices and techniques in graphics ?

15. (a) Describe about raster algorithm.

Or

(b) What are geometrical analyses of structures ? Explain.

Section - C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the derivatives of molecular mechanics energy function.

17. Give a brief account on molecular dynamics with continuous potential energy.

18. Explain brief on treatment of delocalized pi system in molecular mechanics.

19. Write briefly on graphics interaction devices and techniques.

20. Enumerate the method to visualize and represent of molecular structure through computer graphics.

——— *** ———

RW-6080

502506

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

PROGRAMMING IN JAVA AND VB

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.

1. Why do you prefer Java ?

2. Write declaration for two dimensional array of type boolean.

3. Why a java program can not directly communicate with an ODBC driver ?

4. What is an Applet ?

5. How will you create and display image ?

6. Write the purpose of arrays in Java.

7. What is image filters ?

8. What do you mean by Schema ?

9. Explain relational model.

10. What is data control ?

Section - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain in detail about exception handling in Java.

(*Or*)

(b) Explain conditional loops in Java.

12. (a) Explain any six Java event handlers with suitable examples.

(*Or*)

(b) Discuss the states and methods of thread with example.

13. (a) How do Applets differ from Applications ?

(*Or*)

(b) Explain the use of any five Java AWT classes.

14. (a) Explain VB Standard Control.

(*Or*)

(b) Give an account on constants and variables in VB.

15. (a) Briefly discuss working with Code in VB.

(*Or*)

(b) Write notes on Data control in Oracle.

Section - C (3 × 10 = 30)

Answer any **three** questions.

16. Write a Java program to explain the usage of fonts.

17. Develop an applet to draw colored bars and explain.

18. Discuss packages and interfaces in Java.

19. Write an essay on Visual basic Client / Server technology.

20. What do you mean by data connectivity ? Explain the different database approaches with examples.

——— *** ———

RW-6081

502507

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

Elective III—IPR, BIOSAFETY AND BIOETHICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. GATT agreement.

2. G8 summit.

3. Trademark.

4. Geographical indications.

5. Super bug.

6. Licensing.

7. BSL—I.

8. Name two transgenic plants in India.

9. Xenotransplantation.

10. Human genome project.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain the important functions of WTO.

(*Or*)

(b) Explain briefly about Doha level summit.

12. (a) Explain various benefits of TRIPS.

(*Or*)

(b) Write short notes on Geographical indications.

13. (a) Write notes on Patent Act, 1970.

(*Or*)

(b) Explain what are the biological materials can be patented.

14. (a) Write short note on maximum containment laboratory.

(*Or*)

(b) Explain the guidelines for research in transgenic plants.

15. (a) Write short notes on ethical issues related to animal experiments.

(*Or*)

(b) Explain the medical applications of embryonic stem cell cloning.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Write a critical note on the advantages and disadvantages after signing WTO.

17. With suitable example give detailed account on different types of patents.

18. Describe the case studies related to Basmati rice and Turmeric in World Court.

19. Give a detailed account on guidelines for rDNA research in transgenic plants.

20. Write a detailed account on the environmental impacts of transgenic plants and animals.

——— *** ———

RW-6082 **502101**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

MATHEMATICS FOR BIOINFORMATICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** the questions.

1. Find the value of $1 - \sin^2 30°$.

2. If $A = \begin{bmatrix} 5 & 1 \\ 3 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 2 \\ 3 & 4 \end{bmatrix}$,

   calculate $B^2 + AB + 6A + A^2$

3. Find Eigenvalues of a matrix A = $\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$

4. Find the values of a_0 for the Fourier series of

$f(x) = (\pi - x)^2$ in the $(-\pi, \pi)$

5. If A and B are two sets given by A = {1, 3, 5, 7} and B = {5,7, 9, 11, 13,} find A $\cup$ B and A $\cap$ B

6. Calculate tan 60° using the trigonometric formula given that tan 30° = $1/\sqrt{3}$

7. Find the values of equation of the circle passing through (–2, 0) and having its centre at (2, 3).

8. Evaluate $\int ex^2 + \frac{3x}{x^4} + 2x, dx$

9. Find the values of $\cos^2 30° + \sin^2 30$.

10. Find the angle between the two vectors

$\vec{a} = 2\vec{i} + \vec{j} + 2\vec{k}$ and $\vec{b}\ 3\vec{i} + 2\vec{j} + 6\vec{k}$

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Find the values of b_n for the Fourier series of

$f(x) = (\pi - x)^2$ in the $(-\pi, \pi)$.

Or

(b) If $y = \log\left(\frac{1+\sin x}{1-\sin x}\right)$, find $\frac{dy}{dx}$.

12. (a) The position vectors of the vertices of triangle are $\vec{i} + 3\vec{j} + 4\vec{k}$, $3\vec{i} + 4\vec{j} + \vec{k}$, $4\vec{i} + \vec{j} + 3\vec{k}$. Prove that the triangle is an equilateral triangle.

Or

(b) Evaluate $\int 3x^2 e^x \, dx$.

13. (a) Derive the formula for the Laplace transformation of $t \sin at$.

Or

(b) If $\vec{a} = \vec{i} - 2\vec{j} + \vec{k}$, $\vec{b} = 12\vec{i} + 8\vec{j} - 4\vec{k}$, $\vec{c} = 2\vec{i} - 4\vec{j} - 3\vec{k}$,

find $4\vec{a} + \vec{b} - 2\vec{c}$

14. (a) Solve : $\frac{dy}{dx} = 3x^2 + \sin 3x$

Or

(b) Examine the matrix $\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is the inverse of $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

15.(a) If $y = e^{\sqrt{\tan x}}$ find dy/dx

Or

(b) Solve $\sqrt{1-x^2}\ dy + \sqrt{1-y^2}\ dx = 0.$

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Prove that the points with position vectors

$\vec{a} - 2\vec{b} - 3\vec{c}, 2\vec{a} + 3\vec{b} - 4\vec{c}$, and $-7\vec{b} + 10\vec{c}$ are collinear.

17. For the matrix A = $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{bmatrix}$ find the Eigenvectors

18. Find $\frac{dy}{dx}$, $x^y = y^x$

19. Find the equation of the ellipse whose eccentricity is ½, the focus is (–1, 1) and the direction is $x - y + 3 = 0$.

20. $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ find A^{-1} and verify that

$A.A^{-1} = A^{-1}.A$.

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RW-6083 **502203**

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

BIOLOGICAL SEQUENCE ANALYSIS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** the questions.

1. Smith Waterman algorithm.

2. PAM.

3. BLAST.

4. E value.

5. MUSCLE.

6. Cladistics.

7. Motif.

8. ProClass.

9. Average linkage method.

10. Mesquite.

Part - B (5 × 5 = 25)

Answer **all** questions.

11 (a) Give an account on Mutation Data Matrices (MDM).

(*Or*)

(b) Write a note on Block Substitution Matrices.

12. (a) Explain the application of FASTA and its significance.

(*Or*)

(b) Write short notes on Global Alignment.

13 (a) List out the features of MSA.

(*Or*)

(b) Explain the role of Multiple Structural Alignment tool.

14 (a) Give an account on Motif and Domain Databases.

(*Or*)

(b) Give short account on Sequence Logos.

15 (a) Comment on Neighbour joining method and its applications.

(*Or*)

(b) What are the steps involved in constructing a phylogenetic tree ?

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe any two methods of Sequence alignment with suitable examples.

17. Write in detail about different types of BLAST and their significance.

18. Mention the significance and methodology involved in Multiple Alignment.

19. Comment on :

    (a) Position Specific Scoring Matrices.

    (b) Profile analysis.

20. Explain various phylogenetic softwares and their features.

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RW-6084

502503

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioinformatics

Elective II—IMMUNOLOGY FOR BIOINFORMATICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.
All questions carry equal marks.

1. Adaptive immune system.

2. HGPRT.

3. Epitope and Paratope.

4. MHC.

5. IgM.

6. Attenuated vaccines.

7. Dentritic cells.

8. Autoimmune diseases.

9. Multiple sequence alignment.

10. Phylogeny tree.

Part - B (5 × 5 = 25)

Answer **all** questions.

All questions carry equal marks.

11 (a) Describe in detail the types of immune system.

(*Or*)

(b) Write short note on effectors involved in innate immune mechanism.

12. (a) Discuss in detail about different types of antigens that challenge immune system.

(*Or*)

(b) Explain about MHC polymorphism and MHC super types.

13 (a) Describe host pathogen interaction with bioinformatics application.

(*Or*)

(b) Explain different types of vaccines.

14 (a) Give an account on diversity of antibody against antigen recognition.

(*Or*)

(b) Discuss about Infectious diseases and immune system with examples.

15 (a) Explain the types of immune sequence alignments with a sequence alignment tool as reference.

(*Or*)

(b) Discuss how immunological databases used for analyzing Immunoglobulin sequence and structure.

Part - C (3 × 10 = 30)

Answer any **three** of the following.

16. What are immunoglobulins ? Explain its classes and subclasses.

17. Explain hybridoma technology and its applications.

18. Describe different classes of major histocompatibility complex.

19. Write notes on various autoimmune diseases.

20. Write in detail about structure prediction using homology modeling.

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